

Applications of Linear Algebra

Math 235H

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Physics is ultimately the building blocks of all other sciences. What mathematics is to physics: physics is to chemistry, biology, and engineering. This paper will focus on the former—more specifically, what linear algebra does for physics. Linear Algebra consists of a vast and powerful field of information that, when applied to physics, can revolutionize our understanding of it.

Linear Algebra can help significantly simplify very complex systems of equations and, through vectors and matrices, can allow for the modeling of intricate physical systems to help notice or predict patterns in nature. The language of linear algebra is intrinsic to physics as we label almost every value generated with as a "scalar" or a "vector". Matrices are critical to easily solving the countless linear systems generated by physical phenomena. Eigenvalues give meaning to the Hamiltonian matrix. Higher Dimensional Matrices are our way to digest the incredibly complicated tensors that are applied to almost every field in physics.

Linear Systems are critical to the daily functions of life, yet go largely unnoticed when compared to more grandiose applications of linear algebra. The application of linear systems that I am intimately familiar with is Kirchoff's Laws. These laws, when applied to circuits consisting of resistors, capacitors, and inductors, can simplify the complex calculations for voltage, current, and resistance. These three quantities are some of the most important for day-to-day life as they define how the electricity, which defines our lives, actually works.

Kirchoff's first law is that "the current flowing into a node (or a junction) must be equal to the current flowing out of it."¹ Kirchoff's second law is that "in any complete loop within a circuit, the sum of all voltages across components which supply electrical energy (such as cells or generators) must equal the sum of all voltages across the other components in the same loop."¹ These two laws in tandem create a system of linear equations involving three separate "loop currents" as unknowns.

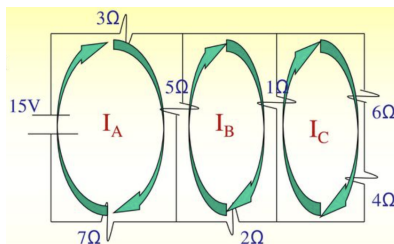


Figure 1: Kirchoff's Law Circuit²

$$\begin{pmatrix} 15 & -5 & 0 \\ -5 & 9 & -1 \\ 0 & -1 & 11 \end{pmatrix} * \begin{pmatrix} I_A \\ I_B \\ I_C \end{pmatrix} = \begin{pmatrix} 15 \\ 0 \\ 0 \end{pmatrix}$$

coefficients unknowns answers

Figure 2: Kirchoff's Law Matrix²

Using the circuit above, we can create the matrix (also above) by considering each "loop" of the circuit to create its own Kirchoff's Law Equations. Using Ohm's Law ($V=IR$), we can easily obtain that:

$$15I_A - 5I_B = 15, \quad -5I_A + 9I_B - 1I_C = 0, \quad -1I_B + 11I_C = 0,$$

which translates easily into the matrix above. This matrix is easily solvable using elementary row operations (another linear algebra topic), and can thus solve for the currents in each loop. This information can then be used in understanding home circuitry, setting up power

Eigenvectors and Eigenvalues can initially seem very redundant or insignificant, however, when observing almost every field, they always have some purpose. The Eigenvalue, in particular, when applied to physics, has a special purpose in regards to the energy spectrum of a system. The Hamiltonian operator is an operator "that represents the total energy of a quantum system, including both kinetic and potential energy."³ This operator acts on "the state space of a quantum system"³, which is an abstract space representing not physical positions, but "quantum states" of the system.

The Eigenvalues of the Hamiltonian "correspond to the possible energy levels of the system"³ and the Eigenvectors of those Eigenvalues "represent the corresponding stationary states"³. If we were to apply the Hamiltonian to Schrödinger's Wave Equation, we can calculate the allowed energy levels and wave functions of a system:

$$i\hbar \frac{d\psi}{dt} = H\psi \text{ where } H\psi = E\psi \text{ and } E \text{ is the total energy of the system.}$$

The Hamiltonian Operator will eventually become $\frac{\hbar^2}{2m} \frac{\partial^2}{\partial t^2} + V(t)$.⁴

Ultimately, with a little more understanding of Quantum Physics, the energy E of the system is equivalent to the "Eigenvalue of ψ when the Hamiltonian acts on ψ ."⁴ This

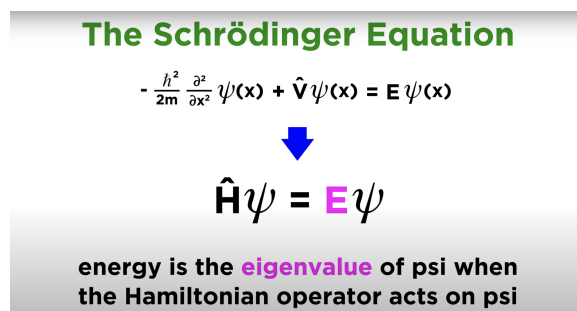


Figure 3: The Conversion with Meaning⁴

incredibly arduous process nets us an understanding of the physical world that we otherwise wouldn't be able to obtain. Although, definitely not applicable to everybody's daily life, the Hamiltonian operator's eigenvalues and eigenvectors are extremely important in the context of quantum mechanics, which fundamentally controls our understanding of planetary motion, molecular behavior, and chaotic systems. Without the Eigenvalues and Eigenvectors, the Hamiltonian couldn't net us the values that control the known universe.

Linear Algebra builds off of itself in increasing magnitudes of difficulty. The most elementary concept of linear algebra are matrices, which allow for easier storage and organization of massive quantities of information. There is another type of matrices, namely "higher-dimensional matrices" that are used in physics extensively. These matrices allow for the understanding and computation of tensors. Tensors, most commonly, are used to describe the "stress in a material object"⁵. Stress is in essence, the amount of force per unit area, typically causing deformation.

$$\begin{bmatrix} \begin{bmatrix} x_{0,0} & x_{0,1} & x_{0,2} \end{bmatrix} \\ \begin{bmatrix} x_{1,0} & x_{1,1} & x_{1,2} \end{bmatrix} \\ \begin{bmatrix} x_{2,0} & x_{2,1} & x_{2,2} \end{bmatrix} \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} \vec{x}_0 \\ \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} \\ \begin{bmatrix} \vec{x}_0 \\ \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} \\ \begin{bmatrix} \vec{x}_0 \\ \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} X_0 \\ X_1 \\ X_2 \end{bmatrix} = \mathcal{X}$$

Figure 4: A "higher-dimensional matrix"⁶

This already complicated matrix of matrices is further complicated by the calculations that have to be done with it in order to net usable information. Although tensors are often unused, their applications go far beyond just theoretical physics. From machine learning to describing the curvature of spacetime, the uses of tensors are innumerable with their understanding rooted in linear algebra.

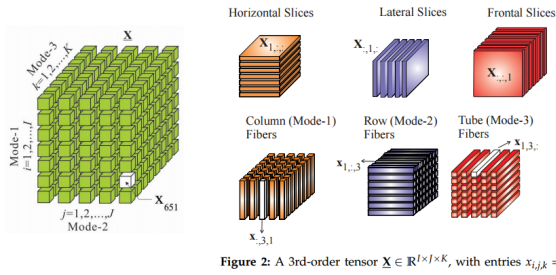


Figure 5: Tensors in Machine Learning⁷

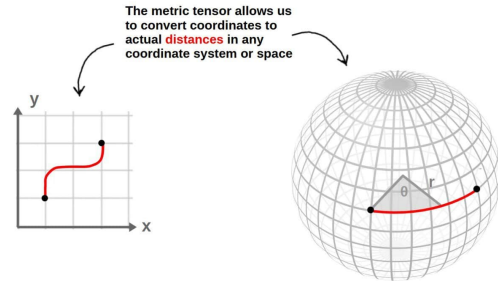


Figure 6: Tensors for Spacetime⁸

In essence, linear algebra as a subject of learning is critical to the inner and outer working of physics. There are countless other applications of linear algebra throughout physics, noticed and unnoticed as linear algebra in it of itself is the basis for almost all higher level studies of any topic. The applications of the various topics within linear algebra is almost limitless, and I personally can't wait to learn about them.

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